

Early Identification of At Risk Students Through Final Mathematics Grade Prediction

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Introduction

Educational institutions gather a great deal of information on students, such as their academic performance, attendance, study habits, family background, and other features relating to the students. By analysing these factors institutions can get a better understanding of student performance and spot those students who might be at risk of having poor academic results.

The project makes use of a machine learning classification technique in order to determine if a student is likely to be classified as At Risk or Not at Risk on the basis of the information available prior to the final grade being assigned. The risk status will be determined by the student's real final Mathematics grade (G3) after a risk threshold has been chosen and properly justified on the basis of the grading context and the results of the exploratory analysis prior to the final model evaluation. The classification model will then learn from the student's earlier academic performance and from other relevant student characteristics in order to predict the student's Risk Status, without using G3 as one of the input features.

The project will make use of the Student Performance dataset available from the UCI Machine Learning Repository, namely the Mathematics dataset (student-mat.csv), with the individual student being taken as the unit of analysis.

The proposed approach is : Student Information + G1 + G2 $\rightarrow$ Classification Model $\rightarrow$ At Risk/ Not at Risk

where:

- G1 is the grade the student received in Mathematics in the first period.
- The grade for Mathematics in the second period is G2.
- G3 is the last grade in Mathematics and it will be used to determine the Risk Status target but will not function as an input feature in the classification model.

The project aims to develop and assess a binary classification model that is able to identify the students who are likely to be at risk of having a poor final Mathematics performance; it will not put into practice any actual academic intervention, the predictions merely serving to provide extra information to academic staff when they make future decisions.

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Selected Track

Selected Track: Guided Data Track

Our group chose the Guided Data Track since the project includes a predefined dataset and enables the group to concentrate on applying machine learning techniques to a real-world educational problem.

The dataset from the UCI on student performance is appropriate for this project since it includes information regarding students' academic records, demographic details, behavioural patterns, and social background, together with their academic grades.

It can be used to find out if a student's previous academic performance and other student characteristics can help in identifying those who are likely to be At Risk.

Our group decided on concentrating on the Mathematics dataset (student-mat.csv), rather than combining the Mathematics and Portuguese datasets. This approach gives them a well-defined and reasonable scope at the same time as enabling them to look into the chosen business problem in sufficient depth.

Business Domain

Domain: Education

The business domain that has been chosen is Education.

Educational institutions ought to monitor students' academic performance and spot the ones who might be at risk of having poor academic results. By identifying students who are likely to be At Risk prior to the final grade being awarded, a machine learning classification model could offer useful information to teachers, academic staff, and the student-support teams.

The project is therefore concerned with using student data in order to aid academic decision-making in a data-driven way.

Business Problem

The main problem addressed by this project is:

In what way can an educational institution make use of a student's previous academic performance and other relevant student characteristics in order to identify those students who are likely to attain poor final Mathematics results?

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When identifying students who are having difficulties academically, members of the academic staff can make use of examination results, observations by teachers, and manual analysis.

A machine learning classification model could provide yet another data-driven approach for spotting students who are likely to end up with poor final Mathematics results before the final grade has been published.

The classification model will predict directly whether or not a student is At Risk, based on the information available prior to the final grade being issued.

It gives academic staff with additional information which could be of use when making future academic decisions.

Business Lenses

For the project, several different business lenses were looked at.

Student Performance Prediction

A first possible lens is Student Performance Prediction.

With this approach, the aim would be to predict a student's final Mathematics grade using information that was available before.

The lens was considered since the dataset includes G1, G2, and G3, which allows one to examine whether a student's earlier academic performance and other student characteristics can be used to predict their final grade.

Reason for considering this lens:

- The dataset has a numerical outcome that is clearly defined, G3.
- G1 and G2 include earlier information on academic performance.
- Regression models allow one to predict the final grade which is numerical.
- The grade that is predicted offers a measurable estimate of how the student is likely to perform academically in the future.

Even though this lens was taken into account it was not chosen since the group's main business objective is to identify students who may be At Risk rather than to predict an exact numerical final grade.

The identification of students at risk – groups' chosen focus

The primary business lens chosen by the group was At Risk Student Identification.

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Using this method, the machine learning model will directly classify the students into two groups: At Risk and Not at Risk. The classification model will base its decision on information that is available prior to the final Mathematics grade being assigned, such as G1, G2, and the relevant student characteristics.

The Risk Status is not an original variable in the UCI dataset; instead it will be calculated from the actual G3 final-grade value by using a risk threshold which will be chosen and properly justified on the basis of the grading context and the results of the exploratory analysis before the final model evaluation. The Risk Status thus obtained will be used as the target variable in the classification model.

The model will thus learn the relationship between a student's previous academic performance and other relevant student characteristics and the student's final risk category. Reason for selecting this lens:

This lens is more relevant in practice to decisions about education than just predicting a numerical grade.

Perhaps the academic staff would rather identify students who are likely to have poor academic results than just know the exact predicted grade. For example:

Student	Risk Status
Student A	Not at Risk
Student B	At Risk
Student C	Not at Risk
Student D	At Risk

This offers a more meaningful interpretation of the model's output from a business point of view.

The project therefore aims to answer:

Which students are likely to have a poor final Mathematics performance?

Identifying students who are at risk might enable members of the academic staff to look at these cases and then consider the appropriate sort of support. Yet the introduction of such interventions is beyond the scope of this project.

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Dataset

The project will make use of the UCI Student Performance dataset – Mathematics dataset (student-mat.csv).

The dataset has details concerning students in secondary education and comprises academic, demographic, social, and behavioural attributes.

Important variables relevant to the proposed task include:

G1 – first-period grade
G2 – second-period grade
G3 – final grade
failures – number of previous class failures
studytime – weekly study time
absences – number of school absences
Medu – mother's education
Fedu – father's education
higher – desire to pursue higher education
schoolsup – extra educational support
famsup – family educational support
age
Other social, behavioural, and demographic characteristics

The Risk Status target will be determined using G3 according to the risk threshold that has been selected in the analysis. G3 will not, however, be included as one of the predictor variables in the classification model.

It is not the case that all the available variables are necessarily included in the final model.

The exact predictor set will not be decided until after the exploratory data analysis, the feature analysis, and it is taken into account whether or not each variable would in fact be available at the time of prediction.

Unit of Analysis

The student is to be regarded as the unit of analysis.

The information about one student is given in each row of the Mathematics dataset.

The model will therefore provide predictions for individual students.

For example:

The student's characteristics along with their earlier academic record are used in order to predict if the student is likely to be classified as At Risk or Not at Risk.

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Proposed Machine Learning Task

The project is going to use a binary classification task in order to achieve the main business aim of identifying students who may be at risk.

The model for classification will make use of information that is available prior to the final Mathematics grade being assigned.

The classification of at risk cases

Input: G1 + G2 + Relevant Student Information

Information about the student that is relevant can include such factors as prior failures, the amount of time spent studying, the number of absences, the educational level of the parents, the amount of educational support received, the student's desire to pursue higher education, and other suitable demographic, social, and behavioural characteristics.

Output: At Risk/ Not at Risk

The Risk Status is not an original variable in the UCI dataset since it will be calculated from the actual G3 final Mathematics grade by using a risk threshold which will be chosen and justified on the basis of the grading context and the results of the exploratory analysis before the final model evaluation.

The process is therefore: $G3 \rightarrow \text{Risk Threshold} \rightarrow \text{Risk Status}$

and the machine learning task is: $\text{Student Information} + G1 + G2 \rightarrow \text{Classification Model} \rightarrow \text{At Risk / Not at Risk}$.

The variable G3 will not be used as an input for the model since it is intended to define the target variable; if G3 were included as a predictor it would result in target leakage and would not reflect a realistic prediction situation.

The model for classification will thus learn if earlier academic performance and other relevant student characteristics can be used to identify those students who are likely to be At Risk before the final grade is released.

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Target Variables and Input Variables

As the project is concerned with At Risk Student Identification, it will make use of one main machine learning objective: Risk Status.

Target Variable: Risk Status – At Risk / Not at Risk

The Risk Status is not an original variable in the UCI Student Performance dataset; it will be calculated from the original G3 final Mathematics grade by using a risk threshold which will be chosen and justified on the basis of the grading context and the results of the exploratory analysis before the final model evaluation.

For instance, if the chosen threshold is $G3 < 10$, then those students who obtain a final grade lower than 10 would be classified as At Risk, whereas students who meet or exceed the threshold would be classified as Not at Risk. The threshold that is finally selected will be one that has been properly justified rather than one that is assumed without any analysis.

Input Variables: The classification model will base itself on the information available prior to the final grade being assigned; the first predictor set might include:

G1 – first-period Mathematics grade
G2 – second-period Mathematics grade
failures
studytime
absences
Medu
Fedu
higher
schoolsup
famsup
age
other relevant student characteristics

The exact predictor set will not be decided until after the exploratory data analysis, the feature analysis, the preprocessing, and it is considered whether or not each variable would actually be available when the prediction is made.

Excluded Variable: G3 will not be used as an input feature. G3 is the last mathematics grade and is used in order to determine the Risk Status target; it would be inappropriate to

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include G3 as an input since the model would have information that directly determines the target it is intended to predict.

Overall Target Structure: The overall structure of the project is therefore: Actual G3 → Risk Threshold → Risk Status

Followed by: Student Information + G1 + G2 → Classification Model → Predicted Risk Status

The project has only one aim in mind when it comes to machine learning, that being the prediction of Risk Status, and only one machine learning task, which is binary classification.

Proposed Model Strategy

The focus of the project will be on binary classification since the primary business objective is to determine whether students are likely to be At Risk or Not at Risk.

Classification Models

A number of classification algorithms will be looked at and compared in order to find out which one gives the best performance for identifying At Risk students.

The proposed models include:

- **Dummy Classifier** offers a simple benchmark whereby the performance of various machine learning models can be compared.
- **Logistic Regression** is used as an easy-to-understand baseline classification model.
- **The Decision Tree Classifier** is used for capturing non-linear relationships and for providing an interpretable tree-type model.
- **The Random Forest Classifier** makes use of a group of decision trees in order to capture more complex relationships.
- **The gradient boosting classifier** is used as an alternative ensemble method with the aim of improving predictive performance.
- **The Support Vector Machine (SVM)** is a method used for margin-based classification.

The same training data and preprocessing method will be used where appropriate when training the models, and their performance will be compared by means of appropriate evaluation metrics.

Classification Evaluation: Accuracy will be taken into account, but it will not be the only factor used to assess the models.

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Particular attention will be given to the following metrics:

- **Recall:** The measure reflects the proportion of genuinely At Risk students who are properly identified; this is significant since failing to identify an At Risk student constitutes an important error with respect to the project's business objective.
- **Precision:** Refers to the proportion of students who are actually classified as At Risk among those who are predicted to be At Risk.
- **F1-score:** Balances Precision and Recall.
- **ROC-AUC:** Measures the model's capability in telling the difference between students who are at risk and those who are not at risk.
- **Confusion Matrix:** Gives the counts of true positives, false positives, true negatives, and false negatives.

Since the main aim of the project is to identify students who may be At Risk, recall will be given particular attention when assessing the model's performance. At the same time, precision will also be taken into account since a model which labels too many students as At Risk could result in a large number of false alarms.

The model to be chosen in the end will be the one that performs best in terms of its predictive ability, gives the most favourable results when validated, and is appropriate for the educational decision-making situation of the project, and not merely the one with the highest level of accuracy.

Validation: The models will be compared using a proper train-test split and stratified cross-validation. This approach will give a more dependable estimate of how the models perform. It also lowers the chance of picking a model based on just one split of the data.

The model selection will not include the test set, which will be used merely for the final evaluation of the chosen model.

Overall Machine Learning Strategy

Firstly: Actual G3 $\rightarrow$ Risk Threshold $\rightarrow$ Risk Status

Secondly: Student Information + G1 + G2 $\rightarrow$ Classification Model $\rightarrow$ Predicted At Risk / Not At Risk

The only time the actual G3 value is used is when preparing the targets in order to obtain the Risk Status label; the risk threshold will be chosen and justified on the basis of the grading context and the exploratory analysis prior to the final model evaluation. The G3 value is not employed as an input feature in the classification model.

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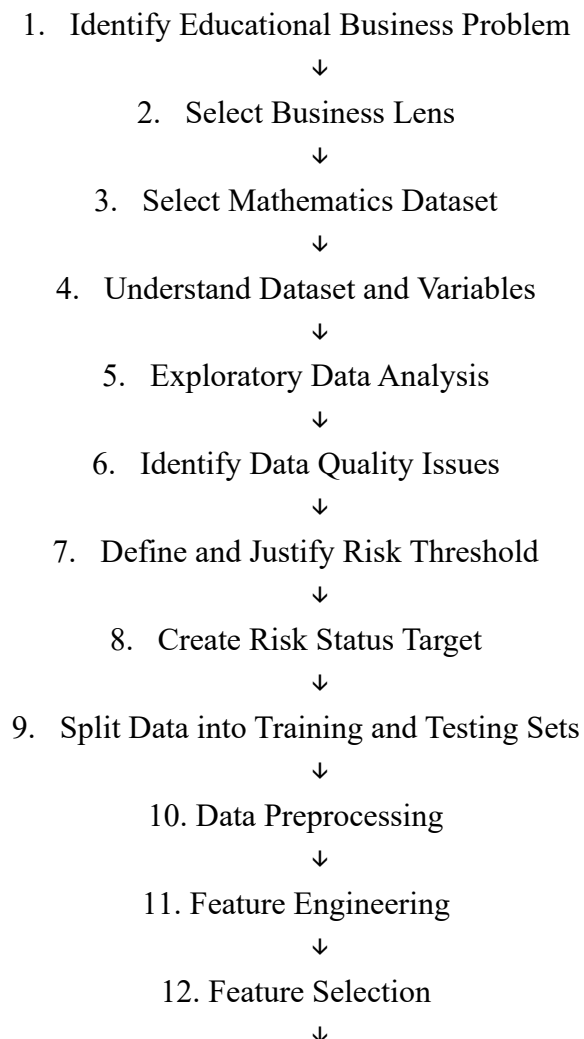
The model for classification will therefore base its learning on the information available prior to the final grade being assigned and will directly state whether or not a student is likely to be At Risk.

This method aligns the machine learning task directly with At Risk Student Identification, the main business objective, and prevents it from involving an unnecessary intermediate prediction of G3.

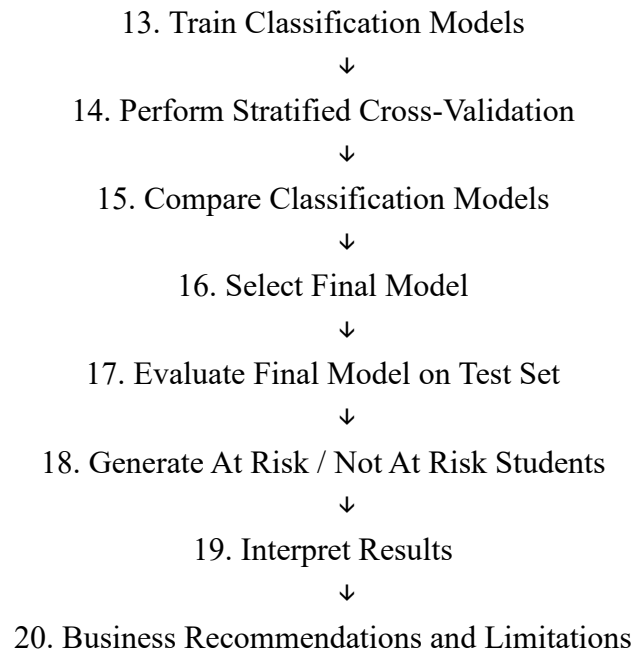
Proposed Project Workflow

The overall project workflow will follow a structured machine learning process, beginning with understanding the educational business problem and dataset and continuing through data preparation, risk-target creation, classification model development, evaluation, and business interpretation.

The proposed workflow:



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Risk Target Preparation

The actual G3 value will be used only during the preparation of the target variable.

The process will be:

Actual G3 → Risk Threshold → Risk Status → At Risk / Not At Risk

The risk threshold will be investigated and justified using the grading context and findings from the exploratory data analysis. Once the risk target has been created, the actual G3 value will not be used as an input feature for the classification models.

Classification Prediction Workflow

The actual machine learning prediction process will be:

Student Information + G1 + G2 → Data Preprocessing → Trained Classification Model → Predicted Risk Status → At Risk / Not At Risk

The classification models will therefore learn from information that would be available before the student's final grade is assigned.

The project will compare several classification algorithms and select the most appropriate model based on its predictive performance, validation results, ability to identify At Risk students and suitability for the educational decision-making context.

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Final Evaluation

After the final classification model has been selected using the training data and stratified cross-validation, it will be evaluated on the previously unseen test set.

The final evaluation will provide an estimate of how well the selected model can classify new students as At Risk or Not At Risk.

The project will conclude with interpretation of the model results, identification of important patterns or predictive factors where appropriate, business recommendations and discussion of limitations and responsible use.

The project ends with prediction and risk identification. Actual academic intervention with students will not be implemented as part of this project.

Core Responsibilities of Group Members

The responsibilities will be split among the four group members according to the steps of the machine learning process.

Member	Core Responsibility
IT24101546	Business problem framing, business lens, stakeholder analysis and Exploratory Data Analysis
IT24100038	Data understanding, data quality assessment, preprocessing, feature engineering and Risk Status target preparation
IT24101024	Classification model development, training, hyperparameter tuning and model comparison
IT24102210	Model evaluation, risk classification analysis, recommendations, limitations and responsible AI

Although each member will have a primary area of responsibility, all members will contribute to the project. Major decisions will be discussed among the group to ensure consistency between the business problem, data preparation, modelling, evaluation, and final recommendations.

The responsibilities are designed so that each major stage of the machine learning workflow is covered while maintaining collaboration between group members.

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IT24101546 – Business Problem and EDA

The first member will be primarily responsible for understanding the educational business problem and investigating the dataset through exploratory data analysis.

The responsibilities will include:

- **Defining the educational business problem:** Clearly establishing the problem of identifying students who may be At Risk of poor final Mathematics performance.
- **Establishing the selected business lens:** Maintaining At Risk Student Identification as the primary business lens and ensuring that the analytical approach is aligned with this objective.
- **Identifying relevant stakeholders:** Considering the needs of academic staff, teachers, student-support teams, and other relevant stakeholders who may use the model's output.
- **Defining the unit of analysis:** Confirm that each individual student is the unit of analysis because every observation is one student.
- **Understanding business requirements:** Identify what information would help decision-making and ensure that the model output is useful from a business perspective.
- **Conducting Exploratory Data Analysis:** Look at the distributions, relationships and patterns in the student data.
- **Investigating relationships between variables and G3:** Examining how G1, G2, and other relevant student characteristics are associated with the final Mathematics grade.
- **Investigating potential At Risk patterns:** Examining whether particular characteristics or previous academic performance patterns are associated with students who may fall below the selected risk threshold.
- **Supporting risk-threshold justification:** Providing EDA findings that can contribute to justify the selected Risk Status threshold.
- **Maintaining the EDA insight log:** Recording important findings and observations identified during the exploratory analysis.

The findings from this stage will support later decisions relating to feature selection, risk-target creation, and interpretation of the classification results.

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IT24100038 – Data Preparation

The second member will be primarily responsible for preparing the dataset for the classification modelling process and ensuring that the data is suitable for analysis.

The responsibilities will include:

- **Understanding the dataset structure:** Examining the rows, columns, observations, and variables contained in the Mathematics dataset.
- **Identifying variable types:** Determining which variables are numerical, categorical, binary, or ordinal and considering the appropriate treatment for each type.
- **Checking data quality:** Investigating the dataset for potential data quality problems that could affect the reliability of the classification models.
- **Investigating missing or inconsistent values:** Identifying missing values, inconsistent entries, or unusual observations and determining appropriate methods for handling them.
- **Defining the Risk Status target:** Using the actual G3 value together with the selected risk threshold to create the At Risk / Not At Risk target variable.
- **Performing the train-test split:** After the Risk Status target has been created, divide the dataset into training and testing sets, using a stratified split where this is appropriate in order to maintain the class distribution.
- **Handling categorical variables:** Applying suitable encoding techniques to categorical variables where required.
- **Applying appropriate preprocessing techniques:** Preparing the features so that they can be used effectively by the selected classification algorithms.
- **Conducting feature engineering where appropriate:** Creating or transforming variables where this can provide useful information for the classification task.
- **Supporting feature selection:** Helping determine which variables should be included as predictors based on their relevance, availability at the prediction stage, and potential contribution to the classification task.
- **Checking for potential data leakage:** Ensuring that information from the final outcome does not incorrectly enter the predictor variables.
- **Documenting preprocessing decisions:** Maintaining clear documentation of the transformations, encoding, feature engineering, and target-generation decisions applied to the dataset.
- **Preparing modelling datasets:** Providing the final training and testing datasets to the machine learning development stage

The actual G3 value will only be used for creating the Risk Status target. It will not be included among the input features used by the classification model.

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IT24101024 – Machine Learning Development

The third member will be primarily responsible for developing, training, and comparing the classification models used to identify At Risk students.

The responsibilities will include:

- **Selecting suitable classification algorithms:** Investigating classification algorithms appropriate for the binary At Risk / Not At Risk prediction problem.
- **Training candidate models:** Train the chosen classification algorithms with the predictor variables.
- **Applying cross-validation:** Use stratified cross-validation to get a reliable estimate of model performance when comparing models.
- **Hyperparameter tuning:** Investigating suitable hyperparameter settings where appropriate to improve model performance.
- **Comparing candidate models:** Comparing the classification models using the selected validation metrics.
- **Investigating feature importance where applicable:** Examining the contribution of predictor variables for models where feature-importance analysis is appropriate.
- **Keeping training procedures for models reproducible:** Make sure that procedures for training, validating, and tuning models are carried out in a consistent manner.
- **Documenting the algorithms:** Keep track of the algorithms used, modelling procedures, approach to cross-validation, model tuning decisions, and comparison outcomes of the models.

The models developed by this member will directly learn the relationship between:

Student Information + G1 + G2

And:

Risk Status – At Risk / Not At Risk

The member will not use the actual G3 value as a predictor and will not create an intermediate predicted G3 value.

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IT24102210 – Model Evaluation and Business Interpretation

The fourth member will be primarily responsible for evaluating the developed classification models and interpreting their results from the perspective of machine learning and the business.

The responsibilities will comprise:

- **Evaluate classification performance:** Assessing the performance of the candidate models using appropriate classification metrics.
- **Analysing Recall:** Pay special attention to Recall for the At Risk class since failing to identify an At Risk student poses an important error for the project's business objective.
- **Analysing Precision:** Considering the proportion of students predicted as At Risk who are genuinely At Risk and the impact of false alarms.
- **Analysing F1 score:** Using F1-score to assess the balance between Precision and Recall.
- **Analysing ROC-AUC:** Evaluating the model's ability to differentiate between At Risk and Not At Risk students across different classification thresholds.
- **Analysing the Confusion Matrix:** Exploring True Positives, False Positives, True Negatives, and False Negatives to comprehend the types of errors made by each model.
- **Selecting the final model:** Supporting the final model-selection decision using the validation results and the project's business requirements.
- **Final test-set evaluation:** Evaluating the selected model on the previously unseen test set after the model selection is complete
- **Interpreting model results:** Interpreting what the final classification results mean in the context of At Risk Student Identification.
- **Developing business recommendations:** Translating the model findings into meaningful recommendations for the educational context.
- **Identifying limitations:** Discussing limitations relating to the dataset, risk threshold, model performance, generalizability, and prediction errors
- **Responsible AI:** Ensuring that model predictions are treated as decision-support information rather than an automatic decision over students.
- **Preparing final conclusions:** Summarizing whether the final classification approach effectively addresses the project's primary business objective.

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The final evaluation will take into account both the statistical performance of the classification model and its practical suitability for identifying students who might need further academic attention.

Expected Project Outcome

The expected outcome of the project is to develop a machine learning-based approach which would be able to identify At Risk and Not At Risk students before assigning the final Mathematics grade, G3.

In particular, the G3 value will be used to define the Risk Status target. An appropriate risk threshold will be selected in order to classify students as At Risk or Not At Risk.

Therefore, the target-generation process can be described as:

Actual G3 → Risk Threshold → Risk Status

Once the Risk Status target has been generated, G3 is excluded from the predictor variables since it is the final outcome that is used to define the Risk Status target. Including G3 as an input would result in target leakage.

Instead, the following information will be used for developing a classification model:

Student Information + G1 + G2

in order to predict:

At Risk/ Not At Risk

Therefore, the complete machine learning pipeline can be described by first estimating:

Actual G3 → Risk Threshold → Risk Status

Followed by:

Student Information + G1 + G2 → Classification Model → Predicted At Risk / Not At Risk

The expected outcome of the project will directly address the problem statement which requires to identify students who might need additional attention from academic staff. In particular, the final model output could serve as an additional indicator for educators in order

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to understand which students might benefit from a personalized intervention program. However, it will be essential to utilize this model as an assistant rather than a decision-maker.

More specifically, the model's output will show if a student is predicted to be At Risk or Not at Risk; this prediction may therefore serve as an extra clue for academic staff as to whether further review or support should be considered.

Initial Limitations

Several limitations are associated with this project, in particular at the initial stages. Some of the most relevant ones are described below.

Dataset Generalizability

The current project uses the UCI Student Performance dataset which only reflects specific circumstances of the students who have participated in the study.

In particular, it can be assumed that an ML model which is trained on this data will only generalize to specific contexts, and not all educational settings.

Dependence on Previous Grades

G1 and G2 are expected to be important predictors because they represent the student's previous academic performance.

Although these variables are available before G3 is assigned and are therefore suitable for the proposed prediction task, the model may rely heavily on previous academic performance rather than other student characteristics.

Risk Threshold Selection

The Risk Status target is created by applying a threshold to the actual G3 value.

Therefore, the definition of At Risk depends on the selected threshold. A different threshold could result in a different number of students being classified as At Risk and could consequently affect the class distribution and model performance.

The threshold must therefore be carefully investigated and justified using the grading context and exploratory data analysis rather than being selected arbitrarily.

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Classification Errors

The classification model may produce both false positives and false negatives.

A false positive occurs when a student who is actually Not At Risk is classified as At Risk.

A false negative occurs when a student who is actually At Risk is classified as Not At Risk.

False negatives are particularly important for this project because an At Risk student may not be identified for further review.

Class Imbalance

Depending on the selected risk threshold, the resulting Risk Status target could have different numbers of At Risk and Not At Risk students.

If large class imbalance is discovered during analysis, some methods should be taken into consideration to ensure that the classification models are not biased toward the majority class.

Prediction Uncertainty

A machine learning classification is a prediction and does not guarantee the actual academic outcome of a student.

The model could make bad predictions due to the student's performance being affected by other variables that were not in the dataset.

Responsible Use

The model should be used to support academic decision-making rather than automatically determining how students are treated.

Students should not be labelled, disadvantaged, or subjected to academic decisions solely on the basis of a machine learning prediction.

The model output should therefore be interpreted together with appropriate human judgement and other relevant information.

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Conclusion

This project proposes a machine learning-based approach for At Risk Student Identification within the Education domain.

Using the UCI Student Performance Mathematics dataset, the project will use information that is available before the final Mathematics grade is assigned, particularly G1, G2, and relevant student characteristics, to identify students who may be At Risk.

The project will use a binary classification approach because the primary business objective is to determine whether a student is likely to be At Risk or Not at Risk.

The actual G3 value will be used only during target preparation. A suitable risk threshold will be applied to the actual G3 value to create the Risk Status label.

The target-generation process is therefore:

Actual G3 → Risk Threshold → Risk Status

After the Risk Status target has been created, G3 will not be used as an input feature in the classification model.

The classification models will instead use:

Student Information + G1 + G2

to directly predict:

At Risk / Not At Risk

As a first step a Dummy Classifier will be used as a simple baseline, and then Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and Support Vector Machine classifiers will be applied. The way in which their performance is evaluated will be by means of appropriate evaluation metrics.

The appropriate evaluation metrics will be used to compare the classification models, including Recall, Precision, F1-score, ROC-AUC, and the Confusion Matrix, with a special focus on Recall for the At Risk class since the main aim of the project is to correctly identify students who may need further academic attention.

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The model selection process will use an appropriate train-test split and stratified cross-validation. The test set will be kept separate from the model selection process and will be used only for the final evaluation of the selected model.

The project will follow a structured workflow covering business problem definition, dataset understanding, exploratory data analysis, data preprocessing, feature engineering, feature selection, risk-target creation, classification modelling, validation, final evaluation, and business interpretation.

The final objective is to provide a data-driven approach for identifying students who may be at risk of poor final Mathematics performance. The resulting predictions could potentially support academic staff in making informed decisions, while actual academic intervention remains outside the scope of the project.

Overall, the proposed approach aligns the machine learning task directly with the project's primary business objective and avoids the need for an unnecessary intermediate prediction of G3.